

ArogyaSetu Bridge — Enterprise Organizational & Engineering Roles Blueprint

Real-World State-Scale Digital Public Health Platform (SIH 2026 | PS ID 26133)

1. Executive Summary & Organizational Strategy

Building and operating a mission-critical, state-scale Digital Public Infrastructure (DPI) for healthcare across Maharashtra's 36 districts (serving 120+ million citizens, 60,000+ frontline health workers, and 3,000+ public facilities) requires an elite, multi-disciplinary engineering and clinical operations team.

To ensure rapid execution, clear ownership, and zero operational ambiguity, the platform utilizes a **14-Role Dedicated Specialist Model** where every discipline has a single, dedicated owner accountable for its architecture, security, clinical validity, and performance.



2. Detailed Roster of 14 Enterprise Roles

01. Chief Solution Architect / CTO (Technical Lead)

- Domain:** Engineering Leadership & System Architecture
- Primary Mandate:** Defines the end-to-end technical strategy, microservice boundaries, technology stack standards, and guarantees 99.99% system availability across the state network.
- Key Codebase Ownership:**
 - `turbo.json`, `package.json` (Root monorepo configuration)
 - `infrastructure/k8s/` (Cluster orchestration blueprint)
 - `packages/shared-types/` (Universal data contracts)
- Tech Stack & Tools:** TypeScript, Node.js, Kubernetes, PostgreSQL, Turborepo, Architecture Decision Records (ADRs).
- Core KPIs:** System uptime (>99.99%), architecture consistency, mean time to recovery (MTTR < 15 mins), technical debt control.

02. Health Informatics Product Manager

- **Domain:** Product Strategy & Public Health Policy Alignment
 - **Primary Mandate:** Bridges state public health policies (NHM, MSInS, IPHS 2022) with software engineering sprints, managing feature backlogs and release roadmaps.
 - **Key Codebase Ownership:**
 - `docs/api/` (API specification governance)
 - Sprint backlogs, user stories, and acceptance criteria
 - **Tech Stack & Tools:** Jira/Linear, Figma, Postman, OpenTelemetry, Confluence.
 - **Core KPIs:** Sprint velocity predictability (>90%), feature adoption rate across pilot PHCs, regulatory milestone delivery.
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03. Medical Director / Clinical Lead (Registered Medical Practitioner)

- **Domain:** Clinical Safety, Triage Protocols & CDSCO Compliance
 - **Primary Mandate:** Validates all clinical decision support rules, oversees the 4-Tier urgency triage matrix, enforces NMC Telemedicine Practice Guidelines (2020), and ensures CDSCO Class B SaMD compliance.
 - **Key Codebase Ownership:**
 - `packages/triage-engine/` (Clinical triage algorithmic rules)
 - `packages/shared-types/clinical.ts` (SNOMED-CT / ICD-10 diagnostic models)
 - **Tech Stack & Tools:** SNOMED-CT browsers, LOINC registries, ICD-10 codings, Clinical Decision Support System (CDSS) validator suites.
 - **Core KPIs:** Triage false-negative rate (<0.1%), clinical protocol accuracy (100%), medical board certification sign-off.
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04. Data Privacy & Cybersecurity Officer

- **Domain:** DPDP Act 2023 Compliance, CERT-In Audit & Cryptography
 - **Primary Mandate:** Enforces statutory data fiduciary obligations under the Digital Personal Data Protection Act 2023 (Sections 8, 9, 10), manages zero-knowledge encryption, consent logs, and CERT-In security audits.
 - **Key Codebase Ownership:**
 - `apps/api-gateway/src/middleware/audit.ts`
 - `apps/api-gateway/src/middleware/auth.ts` (Zero-Trust RBAC)
 - Data protection impact assessments (DPIA) & consent ledgers
 - **Tech Stack & Tools:** AES-256-GCM, PBKDF2, OWASP ZAP, HashiCorp Vault, SonarQube, Wazuh SIEM.
 - **Core KPIs:** Zero security breaches, 100% DPDP Act compliance audit score, annual CERT-In vulnerability certification.
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05. Senior Backend / Distributed Systems Engineer

- **Domain:** Core API Gateway, Microservices & High-Concurrency Pipelines
 - **Primary Mandate:** Builds the central Express.js REST API gateway, microservice controllers, rate-limiting, and background message queues capable of handling 50,000+ concurrent health worker requests.
 - **Key Codebase Ownership:**
 - `apps/api-gateway/` (All route handlers, controllers, and services)
 - `apps/api-gateway/src/services/` (Business logic layer)
 - **Tech Stack & Tools:** Node.js, Express.js, TypeScript, RabbitMQ, BullMQ, Redis, Zod.
 - **Core KPIs:** API p95 response time (<45ms), gateway throughput (>12,000 req/sec), zero unhandled exceptions.
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06. Database Architect & Data Engineer

- **Domain:** Relational Modeling, Indexing, Caching & HL7 FHIR Schemas
- **Primary Mandate:** Manages the 25-table PostgreSQL relational database, schema migrations, composite indexing, connection pooling, Redis caching, and FHIR R4 schema serialization.
- **Key Codebase Ownership:**
 - `database/migrations/` & `database/seeds/`

- `packages/fhir-mapper/` (FHIR R4 resource generators)
 - PostgreSQL stored procedures & Prisma schema models
 - **Tech Stack & Tools:** PostgreSQL 16, pgvector, Prisma ORM, Redis 7, HL7 FHIR R4, `pg_stat_statements`.
 - **Core KPIs:** Query execution p99 (<12ms), zero database deadlocks, database replication lag (<500ms).
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07. Frontend Web Lead (Provider & DHO Portals)

- **Domain:** Desktop Doctor Dashboard & State Health Analytics
 - **Primary Mandate:** Develops the Next.js 14 Provider Portal for PHC doctors, CHC specialists, and District Health Officers (DHOs), implementing real-time OPD queues, e-prescriptions, and epidemiological heatmaps.
 - **Key Codebase Ownership:**
 - `apps/web-dashboard/` (Next.js 14 App Router, Server Components)
 - `packages/ui-components/` (Apple UI Design System React components)
 - **Tech Stack & Tools:** Next.js 14, React 19, TypeScript, TailwindCSS/Vanilla CSS, TanStack Query, Recharts/D3.js.
 - **Core KPIs:** First Contentful Paint (<0.6s), Lighthouse Accessibility score (100), doctor encounter entry time (<20 seconds).
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08. Mobile & Edge Engineer (Frontline Worker PWA)

- **Domain:** Offline-First ASHA/ANM Application & Device Ergonomics
 - **Primary Mandate:** Builds the ultra-lightweight, offline-first field application for 60,000+ ASHA workers, ensuring seamless operation on entry-level 2GB Android Go devices with low battery drain.
 - **Key Codebase Ownership:**
 - `apps/mobile-app/` (React Native / Progressive Web App)
 - `packages/sync-engine/` (Local IndexedDB / Dexie.js persistence layer)
 - **Tech Stack & Tools:** React Native, Expo, Dexie.js, Workbox Service Workers, Web Storage API.
 - **Core KPIs:** Application bundle size (<8 MB), RAM footprint (<140 MB), battery consumption (<4% per 8-hour shift).
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09. Real-Time Video/Audio Communications Engineer

- **Domain:** Teleconsultation Media Servers & Bandwidth Adaptation
 - **Primary Mandate:** Architects the low-latency WebRTC video and audio teleconsultation pipeline, STUN/TURN traversal servers, and adaptive bitrate streaming for rural 2G/3G connections.
 - **Key Codebase Ownership:**
 - `apps/api-gateway/src/controllers/teleconsult.controller.ts`
 - `infrastructure/docker/coturn/` (TURN/STUN server configuration)
 - **Tech Stack & Tools:** WebRTC, mediasoup, Coturn, Opus Audio Codec, VP8/H.264, Web Audio API.
 - **Core KPIs:** Audio packet loss concealment (>95% intelligibility on 2G), video connection success rate (>98.5%), audio consent recording integrity (100%).
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10. GovTech & ABDM Integration Engineer

- **Domain:** National Health Authority (NHA) & State Government Bridges
 - **Primary Mandate:** Implements direct API integrations with the ABDM Gateway (Milestones M1 ABHA, M2 Health Data Linking, M3 Consent Manager), e-Aushadhi / DVDMS state warehouse, and MJPJAY insurance systems.
 - **Key Codebase Ownership:**
 - `apps/api-gateway/src/routes/abdm.routes.ts`
 - `apps/api-gateway/src/routes/eaushadhi.routes.ts`
 - `packages/eaushadhi-bridge/`
 - **Tech Stack & Tools:** ABDM Sandbox APIs, OAuth 2.0 Mutual TLS, REST/SOAP wrappers, RSA/ECC signature verifiers.
 - **Core KPIs:** ABDM M1/M2/M3 milestone certification (100%), e-Aushadhi daily stock synchronization success (>99.9%).
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11. Edge Mesh & Embedded Sync Engineer

- **Domain:** Peer-to-Peer Zero-Connectivity Mesh Protocols
 - **Primary Mandate:** Develops peer-to-peer Bluetooth Low Energy (BLE) and Wi-Fi Direct synchronization protocols using Concise Binary Object Representation (CBOR) to sync health records across media-dark tribal villages.
 - **Key Codebase Ownership:**
 - `packages/mesh-sync/` (BLE discovery, socket pairing, and delta sync)
 - `packages/sync-engine/cbor-codec.ts` (Compact serialization)
 - **Tech Stack & Tools:** Web Bluetooth API, Wi-Fi Direct APIs, CBOR (RFC 8949), Conflict-Free Replicated Data Types (CRDTs).
 - **Core KPIs:** Compressed clinical payload (<4.2 KB per encounter), offline P2P sync completion rate (>99.8%), zero data loss during sync.
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12. UI/UX & Vernacular Accessibility Designer

- **Domain:** Low-Literacy Interface Design & Vernacular Usability
 - **Primary Mandate:** Creates the Apple UI / VisionOS-inspired design system tailored for rural health workers, featuring large tap targets, intuitive clinical color-coding, and vernacular dialect flows (Marathi, Ahirani, Varhadi, Gondi).
 - **Key Codebase Ownership:**
 - Design Tokens (`--brand-orange` , `--brand-blue` , glassmorphic styles)
 - Iconography sets for symptom screening & vernacular microcopy
 - **Tech Stack & Tools:** Figma, Apple Human Interface Guidelines (HIG), Web Accessibility (WCAG 2.1 AAA).
 - **Core KPIs:** ASHA task completion without assistance (>94%), user error rate (<2%), user interface satisfaction score (>4.8/5).
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13. Cloud DevOps & Site Reliability Engineer (SRE)

- **Domain:** Infrastructure as Code, CI/CD & MeghRaj/NIC Cloud Hosting
 - **Primary Mandate:** Manages the sovereign cloud infrastructure (NIC MeghRaj / AWS India), automated CI/CD deployment pipelines, zero-downtime rolling updates, and 24/7 telemetry monitoring.
 - **Key Codebase Ownership:**
 - `.github/workflows/` (CI/CD test & deploy pipelines)
 - `infrastructure/k8s/` (Kubernetes deployments, ingress, and secrets)
 - `infrastructure/monitoring/` (Prometheus & Grafana dashboards)
 - **Tech Stack & Tools:** Kubernetes (K3s/EKS), Terraform, Docker, GitHub Actions, Prometheus, Grafana, Loki.
 - **Core KPIs:** Deployment frequency (multiple per week with zero downtime), infrastructure cost efficiency, incident MTTR (<15 mins).
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14. Lead QA & Security Test Engineer

- **Domain:** Automated Testing, k6 Load Simulations & Penetration Auditing
 - **Primary Mandate:** Builds end-to-end automated testing suites, tests edge-case data sync conflicts, executes high-volume load simulations (100,000 virtual users), and validates clinical data integrity.
 - **Key Codebase Ownership:**
 - `tests/e2e/` (Playwright / Cypress automated test suites)
 - `tests/load/` (k6 stress and soak testing scripts)
 - **Tech Stack & Tools:** Playwright, k6, Jest, Supertest, OWASP ZAP, Artillery.
 - **Core KPIs:** Automated test code coverage (>88%), load test tolerance (50,000 peak concurrent users with zero 5xx errors).
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3. RACI Responsibility Assignment Matrix

Platform Module	Architect (01)	Product (02)	Medical (03)	Security (04)	Backend (05)	DB (06)	Web (07)	Mobile (08)	Media (09)	GovTech (10)	Mesh (11)	UI/UX (12)	DevOps (13)	QA (14)
4-Tier Triage Engine	A	C	R	C	R	C	C	R	I	I	I	C	I	R
25-Table DB & FHIR	A	I	C	C	C	R	I	I	I	C	I	I	C	R
WebRTC Teleconsult	A	C	C	C	C	I	R	R	R	I	I	C	C	R
ABDM M1/M2/M3 Bridge	A	C	I	C	C	C	I	I	I	R	I	I	C	R
Offline P2P Mesh Sync	A	C	I	C	C	C	I	R	I	I	R	I	C	R
ASHA Incentive Ledger	C	R	C	I	R	C	I	R	I	C	I	C	I	R
e-Aushadhi / FEFO	C	C	C	I	R	R	C	I	I	R	I	I	I	R
Cloud Hosting & SRE	A	I	I	C	I	I	I	I	I	I	I	I	R	R

Legend: R = Responsible for execution, A = Accountable / Sign-off authority, C = Consulted, I = Informed.

4. Project Delivery & Team Synergy

This 14-role structure ensures that every single requirement of Smart India Hackathon Problem Statement 26133—from clinical safety and tribal dialects to zero-connectivity sync and sovereign cloud deployment—is covered by a world-class dedicated professional.